Are Self-Evolving Agent Skills Learning Skills?

Anonymous submission

Abstract

We introduce a lifecycle audit for determining whether score gains from self-evolving agent skills represent retained skill improvement. A post-revision score alone is insufficient because optimizer variation, validation selection, execution randomness, and verifier choice can improve the endpoint without creating a durable artifact effect. On 198 task variants, SkillEvolver yields a 14.90-point paired gain over no skill and outperforms SPARK and CoEvoSkills, including on the subset where all three supply artifacts. Across SkillRevise, SkillHone, and SkillOpt, we combine frozen parent-to-final deployment, controlled task variation, feedback and selection interventions, artifact-component mediation, and output-locked verifier counterfactuals. The audit separates five evidential claims: selected gain, retained artifact effect, bounded transfer, evolution-pathway dependence, and measurement dependence. This claim-grounded framework lets self-evolving systems report what their experiments establish without equating a better benchmark score with reusable skill learning.

Introduction

Persistent agent skills offer an appealing route to continual improvement. Instead of updating model parameters, an agent can revise an external artifact that stores reusable instructions, scripts, references, templates, or other supporting resources. Recent systems generate, repair, select, and retain such artifacts from execution traces and evaluator feedback, often reporting higher post-evolution benchmark scores as evidence that the agent has learned a better skill (Li and Hu 2026; Liu et al. 2026b; Yang et al. 2026; Xia et al. 2026).

A higher score is not, by itself, an identified skill effect. The selected run may be unusually favorable; the optimizer may produce variable candidates; a validation gate may retain a narrow patch; repeated execution may eventually succeed without a durable behavioral change; and a verifier may reward only part of the intended behavior. These possibilities do not imply that self-evolution is ineffective. They imply that an endpoint score alone cannot locate where the improvement resides.

We therefore ask two connected questions. When self-evolution raises the benchmark score, does the revised skill itself retain a behavioral improvement under fresh deployment and controlled task variation? Regardless of whether it does, which parts of the evolution-and-evaluation lifecycle produced the observed gain? We study these questions in SkillRevise, SkillHone, and SkillOpt, which represent trace-conditioned revision, history-based iterative honing, and validation-gated bounded editing. They form a mechanism-stratified sample of frozen-agent, feedback-driven systems with persistent external skill artifacts, rather than an exhaustive ranking of self-evolution methods.

We first evaluate SkillEvolver, SPARK, and CoEvoSkills on standard tasks and controlled variants, then use artifact content and execution traces to examine how successful procedures adapt to changed inputs and requirements.

Our audit follows the lifecycle from feedback to revision, selection or adoption, frozen artifact deployment, and measurement. We separate historical success from fresh retention, final-skill utility from actual parent-to-final revision effects, and same-task retention from controlled transfer. We then intervene on feedback, optimizer sampling, gate provenance, artifact components, and verifier choice. A short information-integrity check bounds direct answer or oracle exposure as an alternative explanation; it is not the paper's organizing hypothesis.

The resulting picture is deliberately more qualified than either *self-evolution works* or *self-evolution only exploits the evaluator*. Revised artifacts can cause real behavioral improvements, but the demonstrated effects are often narrow. Single-run or selected gains need not survive fresh deployment. Useful traces and feedback must be converted into an effective edit, adopted into the persistent artifact, and retained behaviorally. Finally, a better score can reflect compatibility with a measurement surface rather than broader task behavior.

Our central contribution is a claim-grounded lifecycle audit: each reported gain is matched to the intervention required to support it and to the strongest conclusion that intervention permits. We make three contributions:

- **Identification framework.** We separate a reported gain from retention, behavioral scope, evolution-and-selection pathway, and measurement dependence.
- **Lifecycle audit.** We apply matched interventions across three representative skill-evolution mechanisms without turning heterogeneous studies into a method leaderboard.

This is an anonymized submission for review purposes only. Distribution, citation, or public sharing of this manuscript is strictly prohibited. Copyright and publication details will appear in the final version if accepted.

- **Reporting protocol.** We specify the lineage, contrasts, and artifacts required to support each claim level, from selected gain to bounded transfer.

Problem Formulation

Skill Revision and Evaluation

At revision round r, an optimizer updates the current persistent skill artifact:

$$s_{r+1} = O(s_r, \tau_r, f_r), \tag{1}$$

where s_r is the current skill, τ_r is an execution trace, and f_r is feedback that may include a verifier result, a gold answer, a diagnosis, or evaluator feedback. The operator O denotes the method's complete update pathway, including its native merge, validation, selection, or adoption mechanism. A generated candidate that is rejected or never adopted is not s_{r+1}.

The agent then executes task x with the frozen skill:

$$o_r = A(x, s_r, z_r), \tag{2}$$

where z_r represents model sampling, tool use, and environment state, and o_r is the resulting output artifact or state change. A verifier maps the output to a score:

$$y_r = V(x, o_r). \tag{3}$$

The Identification Question

Most evaluations observe:

$$y_{r+1} > y_r. \tag{4}$$

This inequality describes two evaluations; it does not identify what the persistent skill learned. The revised artifact may encode a retained behavioral improvement, the selected score may depend on optimizer or execution variation, or the output may better match the verifier's measurement surface. Infrastructure-invalid observations, such as a verifier that never reaches its semantic assertions, are neither successes nor failure mechanisms and must be repaired or excluded.

Accordingly, we separate two questions:

1. **Effect existence and scope:** Does the revised skill retain a behavioral improvement under fresh deployment, and how far does that effect extend under controlled task variation?
2. **Effect attribution:** Which feedback, revision, selection, artifact, execution, and measurement mechanisms produced the observed score change?

The following experiments address these questions in turn.

Experiments on SkillsBench

Experimental Setup

We evaluate SkillEvolver (Zhang et al. 2026a), SPARK (Zhou et al. 2026b), and CoEvoSkills (Zhang et al. 2026b) on SkillsBench (Li et al. 2026), with two runs per method. Each run pairs an execution using the generated or evolved skill with a no-skill execution of the same task. Both runs of each method use the same task set: 81 tasks for SkillEvolver, 84 for SPARK, and 85 for CoEvoSkills.

All six runs use Codex with GPT-5.5 and medium reasoning effort. SkillEvolver uses the ChatGPT subscription channel in both runs, SPARK uses the OpenLux API in both runs, and CoEvoSkills uses the subscription channel in run 1 and OpenLux in run 2. Each method follows its own skill-generation procedure. SPARK produces skill artifacts for 43/84 and 41/84 tasks; tasks without a generated artifact remain in the evaluation as no-skill executions. For CoEvoSkills run 2, duplicate evaluation records are resolved after collection by selecting the latest completed valid pair for each task, preserving the baseline and evolved results from the same pair.

We report mean verifier reward on a 0–100 scale and the mean within-task difference between the skill and no-skill conditions. For a run with N tasks, the paired gain is

$$\Delta = \frac{100}{N} \sum_{i=1}^{N} \left(r_i^{\text{skill}} - r_i^{\text{no-skill}} \right). \tag{5}$$

We also count tasks with positive, negative, and zero differences.

Results across Two Runs

All three methods achieve positive mean paired gains in both runs (Table 1). SkillEvolver improves from 30.20 to 55.81 in run 1 and from 37.32 to 62.34 in run 2, yielding gains of 25.61 and 25.02 percentage points. The corresponding task counts are 27 improvements versus six regressions, and 26 improvements versus three regressions.

SPARK increases mean reward from 17.82 to 19.66 in run 1 and from 14.37 to 20.81 in run 2. Its paired gain rises from 1.84 to 6.45 points, with nine versus 13 improved tasks and eight versus five regressions. CoEvoSkills improves from 42.21 to 52.44 and from 23.54 to 33.31 on the aligned 85-task set. Its paired gains are 10.24 and 9.76 points, with 17 improved tasks in each run and nine versus ten regressions.

The direction of the average gain repeats for every method. SkillEvolver's gain remains near 25 points and CoEvoSkills' near ten, while SPARK shows a larger change in gain between runs. Improvements and regressions coexist within each run, making the paired task outcomes informative alongside the aggregate score. The benchmark results quantify the utility of generated and evolved skills under paired deployment. The variant experiments below examine how these gains extend to changed tasks and which artifact-level decisions support transfer.

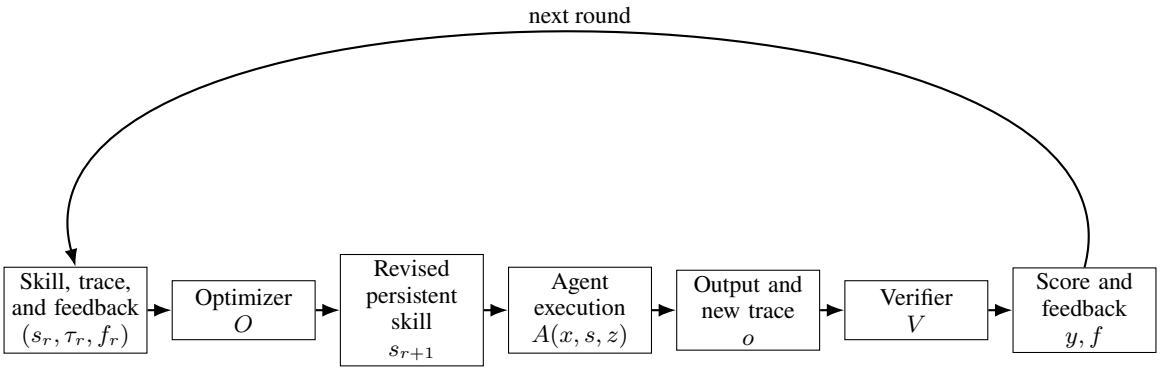


Figure 1: Minimal self-evolution lifecycle. The update operator includes any method-native validation, selection, merge, or adoption before the revised skill is retained.

Method	Run	Tasks	No skill	Skill	Δ	Improved	Regressed	Tied
SkillEvolver	1	81	30.20	55.81	+25.61	27	6	48
SkillEvolver	2	81	37.32	62.34	+25.02	26	3	52
SPARK	1	84	17.82	19.66	+1.84	9	8	67
SPARK	2	84	14.37	20.81	+6.45	13	5	66
CoEvoSkills	1	85	42.21	52.44	+10.24	17	9	59
CoEvoSkills	2	85	23.54	33.31	+9.76	17	10	58

Table 1: Two runs per method on SkillsBench. Scores are mean verifier rewards multiplied by 100; Δ is the mean paired gain in percentage points. Each method uses an identical task set across its two runs. SPARK run 1 denotes the Rep01 budget rerun, and run 2 denotes Rep02.

Generalization under Task Variations

We test how source-task skills perform when the task changes along three axes: input representation (P1), task-instance structure (P2), and coupled task requirements (P3). The reviewed collection contains 222 variants from 77 task families. Complete, unambiguous historical scores for all four conditions define a common comparison set of 198 variants from 71 families. Each variant contributes a no-skill score and scores for SkillEvolver, SPARK, and CoEvoSkills. The separate recovery evaluations are retained with their own protocol records. Table 2 summarizes paired outcomes within the historical comparison set.

SkillEvolver retains a substantial advantage on the variants: mean reward rises from 41.90 without a skill to 56.80, a paired gain of 14.90 points. It improves 46 variants and regresses on 16, compared with 29 improvements and 32 regressions for SPARK, and 25 improvements and 34 regressions for CoEvoSkills. SkillEvolver's paired gains are positive in all three variant groups: 14.89 points on P1, 17.03 on P2, and 12.85 on P3. The structural P2 group shows the largest separation: SPARK and CoEvoSkills have paired changes of -2.43 and -6.98 points, respectively.

Artifact availability differs across methods. SPARK has no generated skill for 95 of the 198 variants, whereas SkillEvolver and CoEvoSkills supply artifacts for all 198. On the 103 variants where every method supplies a skill, SkillEvolver still reaches 69.87, compared with 57.17 for SPARK and 51.46 for CoEvoSkills. Its paired advantages over these methods are 12.70 points (95% CI $[1.97, 24.35]$) and 18.42 points ($[6.68, 30.36]$). These percentile intervals use 20,000 bootstrap samples of the 36 base-task families, retaining each sampled family's variants together. The advantage therefore persists when the comparison is restricted to supplied artifacts.

Tracing the Source of Transfer

Execution traces connect the aggregate result to concrete decisions encoded in the skills. In exoplanet period detection, SkillEvolver separates candidate discovery from refinement and checks whether a dominant peak reflects stellar modulation or a transit. After the input columns are reordered in P1, the executor rejects the approximately 1.395-day modulation and recovers the accepted 5.35907-day period. When P3 requires two signals, the same frozen skill supports a narrower-duration search and refinement to 2.40052 and 5.35958 days. CoEvoSkills' end-to-end period routine encounters duration-grid errors in P1 and returns the stellar-modulation period; in P3 it selects approximately 1.40 days and its 0.70-day harmonic as separate signals. SkillEvolver succeeds on P1 and P3, while CoEvoSkills fails both. On the replacement-data P2 instance, both skills receive zero, giving the case a specific transfer pattern across the three changes.

Method	Variants	No skill	Skill	Δ	Improved	Regressed	Tied
A. Common historical comparison set: 71 task families							
SkillEvolver	198	41.90	56.80	+14.90	46	16	136
SPARK	198	41.90	43.63	+1.73	29	32	137
CoEvoSkills	198	41.90	39.34	−2.56	25	34	139
B. All three methods supply a skill: 36 task families							
SkillEvolver	103	51.21	69.87	+18.67	25	6	72
SPARK	103	51.21	57.17	+5.96	18	14	71
CoEvoSkills	103	51.21	51.46	+0.25	16	15	72

Table 2: Descriptive comparison on matched task variants. Scores and paired gains use the same scale as Table 1. Panel B restricts Panel A to variants whose launch records confirm an available skill for every method, separating artifact availability from performance when artifacts are supplied.

Scheduling locates the difference at a constraint that determines correctness. SkillEvolver's artifact explicitly preserves baseline-relative right shifts and precedence-aware local minimality. The executor carries these rules across renamed inputs, a replacement scheduling instance, and an added machine-handoff requirement, passing all three variants; SPARK also passes all three. CoEvoSkills passes P3, but its P1 output fails the local-minimality check: an operation starts at 29 with an anchor of 24 and can move earlier. The other structural and policy checks pass. This trace-based attribution connects the outcome to preservation of a repair invariant. Together, the cases illustrate a transferable skill as a set of diagnostic and constraint-preserving decisions that the executor adapts to the new task.

Identifying Retained Skill Effects

Evaluation Design and Contrasts

We analyze three persistent-skill systems with distinct revision mechanisms. SkillRevise converts execution traces and diagnoses into bounded repair principles (Liu et al. 2026b). SkillHone accumulates practice history and iteratively reflects on the persistent skill (Li and Hu 2026). SkillOpt produces structured bounded edits and uses a validation gate and rejected buffer before adoption (Yang et al. 2026). The systems are studied as representative mechanisms, not as directly comparable leaderboard entries.

At the paper level, the evidence is exploratory: individual interventions use frozen inputs and predeclared gates, but the combined studies are designed to identify candidate mechanisms rather than estimate benchmark-wide prevalence. A targeted 24-task SkillsBench panel tests whether historical outcomes survive fresh deployment. The panel is stratified to expose stability failures and method disagreements; it is not a random sample. Three controlled mechanism cases then compare complete parent and final artifacts under predeclared variations. Within each contrast, we hold the executor model, harness, task instance, budget, and verifier version fixed unless one is the named intervention; exact prompts, hashes, seeds, and stopping decisions are preserved in the artifact package. The task is the primary analysis unit, while repeated runs are deployments of the same task-condition pair.

Four comparisons must remain distinct. Historical-pass to fresh-repeat comparisons test retention of a selected outcome, not advantage over a baseline. Final-to-no-skill comparisons estimate fresh utility, not the effect of revision. Only actual-parent to final comparisons identify a revision effect. Applying that comparison to controlled variants tests bounded transfer, not population-average benefit.

Panel and contrast	Main result	Supported inference and boundary
Panel A: Retention under fresh deployment		
Historical pass → fresh repeats	12/24 tasks flip within at least one frozen condition; selected-outcome retention is 47.1% for SkillRevise and 55.0% for SkillHone	A selected pass is not a deployment probability; denominators differ and are not a method ranking.
SkillRevise final → no-skill	16/48 vs. 14/48; $+4.2$ pp, 95% CI $[-10.4, 20.8]$	No clear fresh-utility advantage on this targeted panel; not a parent-to-final revision estimate.
SkillHone final → no-skill	12/48 vs. 14/48; -4.2 pp, 95% CI $[-20.8, 14.6]$	No clear fresh-utility advantage on this targeted panel; not a parent-to-final revision estimate.
Panel B: Revision effects under controlled variations		
Invoice parent → final	$0/12 \rightarrow 12/12$; three new same-contract corpora: $0/9 \rightarrow 9/9$	Stable but narrow same-contract transfer.
Energy parent → final	Original task plus 5% parameter variant: $1/6 \rightarrow 5/6$	Directional parameter transfer in a small stochastic case.
SkillOpt SearchQA parent → final	Independent held-out $9/24 \rightarrow 17/24$; stable-transfer gate fails	Local held-out gain, not stable general procedure learning.

Table 3: Evidence for retained skill effects. Panel A contains 24 targeted tasks, three conditions, and two fresh deployments per condition (144/144 behavior-valid runs). Different historical-retention denominators preclude horizontal method ranking. The no-skill condition calibrates execution and selection variability; final-versus-no-skill does not identify revision effects.

Retention under Fresh Deployment

This experiment tests whether historically selected outcomes survive when the final artifact is frozen and redeployed. Holding task, skill, and condition fixed does not make execution deterministic. Across the 24-task panel, 12 tasks change success status between their two repeats in at least one condition. Aggregate fresh success is 14/48 without a skill, 16/48 with the SkillRevise final artifact, and 12/48 with the SkillHone final artifact. Both matched-lift intervals cross zero (Table 3).

These results establish an identification failure in the common reporting practice: cached or selected success cannot be read as stable deployment capability. They do not establish that either revision method is ineffective. The panel is purposive, and it lacks each method's actual parent condition. Its proper conclusion is narrower: on this sample, neither final-versus-no-skill comparison establishes a clear average fresh-deployment advantage, and fresh repeats are necessary before a selected score is called retained.

Revision Effects under Controlled Variations

These mechanism probes test whether the actual parent-to-final revision effect persists under predeclared task changes. In Invoice, the complete final skill repairs all 12 matched deployments and retains the effect on nine deployments drawn from three disjoint corpora that preserve the contract while changing the documents. This is strong evidence of a stable, same-contract revision effect, but not of broad cross-contract generalization.

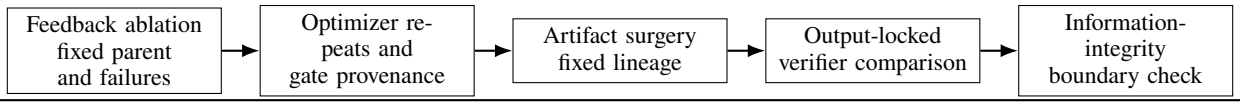
The Energy case provides weaker but directionally consistent evidence: the complete final artifact improves the original setting and a 5% parameter variant in five of six deployments, compared with one of six for the parent. The sample is too small to support a prevalence or generality claim. In SkillOpt SearchQA, the final artifact improves independent held-out accuracy from 9/24 to 17/24, yet fails the predeclared stability threshold. It therefore supports local held-out benefit rather than stable transferable procedure learning.

Taken together, revised skill artifacts can produce retained behavioral effects, but a selected benchmark pass alone does not identify such an effect, and the effects established so far remain narrow rather than broadly generalizable.

Attributing Observed Skill Gains

Lifecycle Interventions

The experiments in this section intervene at different lifecycle stages. They do not estimate one common effect size. Each intervention changes one object while holding its relevant lineage fixed: feedback-view studies hold the parent and failures fixed; optimizer-repeat studies hold the input evidence fixed; gate audits preserve accepted and rejected candidates; artifact surgery holds the surrounding artifact fixed; and verifier counterfactuals hold the output fixed.



Lifecycle stage	SkillRevise	SkillHone	SkillOpt
Feedback and revision	Diagnosis ablation; revision traces	Feedback-view ablation	Feedback branches; optimizer repeats
Selection and adoption	Revision history	Iterative honing history	Native validation gate; accepted and rejected candidates
Artifact mediation	Not yet matched at the same strength	Invoice and Energy surgery	SearchQA component surgery
Measurement	No output-locked comparison yet	Radar verifier swap	Spreadsheet dual scoring
Information integrity	Artifact forensic	Artifact forensic	Wrong-metadata and oracle controls

Figure 2: Lifecycle attribution design and current evidence coverage. The upper panel locates each intervention; the lower panel records method coverage. Populated cells do not imply equivalent strength, and missing cells bound cross-method claims.

Figure 2 makes the current coverage and its gaps explicit. The shared audit logic is that feedback access, candidate revision, validation acceptance, adoption into the persistent artifact, and retained behavioral improvement are distinct events.

Revision and Selection Pathways

These interventions test whether the observed endpoint depends on the feedback view, optimizer sample, or selection gate. In SkillRevise, the complete three-revision pipeline succeeds on 53/86 tasks, compared with 28/86 when diagnosis is removed. Task-level discordance is 26 gains versus one loss, and 2,326 of 3,648 revision entries explicitly use verifier evidence. This demonstrates that diagnosis changes the selected endpoint and that the revision process is strongly evaluator-facing. It does not, by itself, establish which changes are retained under fresh deployment.

SkillHone separates access to feedback from stable benefit. Detailed sanitized, binary, and redacted-tail feedback obtain first-deployment success on 4/12, 4/12, and 5/12 cases, while four of 18 repeated cells flip. Feedback can trigger revisions, but these data do not support a stable advantage for one feedback view.

SkillOpt further separates candidate production from attribution and adoption. In matched SearchQA optimizer repeats, raw-gold and outcome-only views each produce four applied edits. Yet the preregistered between-condition effect size is only $D = .044 < .20$, despite weak lexical decodability, so a single patch cannot be cleanly attributed to feedback view. Across outcome-only and trajectory-only branches, all 6/6 candidates change and 3/6 pass the gate, but every acceptance is driven by the same Blackwater validation item. In OfficeQA, a selected one-item gain disappears in fresh matched deployment. Thus edit production, gate acceptance, and retained improvement cannot be collapsed into one event.

Artifact-Level Mediation

This experiment tests whether a feedback-derived component, rather than unrelated edits in the full artifact, causes the behavior after adoption. Let P denote the complete parent, F the complete final artifact, $P + B$ the parent with component B, and $F - B$ the final artifact with that component removed.

Case	Result	Maximum claim
Invoice	$P = 0/12$, $P + B = 12/12$, $F - B = 0/12$, $F = 12/12$	B is necessary and sufficient for the narrow same-contract effect.
Energy	Local mediation in the original task; necessity and sufficiency do not survive the variant	Local component effect, not transferable mediation.
SearchQA	9 repairs, 0 regressions; stable success 7/12, below the strong gate	Directional mediation on part of the held-out set.

Table 4: Artifact-level mediation evidence.

These interventions establish that an adopted artifact component can cause behavior. They do not show that the same mechanism is prevalent across methods. In particular, SkillRevise lacks a matched surgery of equal strength, and the current cases should be read as existence and boundary evidence.

Measurement Dependence under Fixed Outputs

This counterfactual tests whether a verdict changes when the output is fixed and only the measurement changes. In the Radar family, all 12 reconstructed outputs reproduce their historical original-verifier outcomes. Of six original passes, only one remains a pass under a reference-calibrated strict verifier; the five valid flips arise from specification-implied checks. The same behavior can therefore receive different valid verdicts because the measurement surfaces differ.

Measurement dependence can also operate in the other direction. For identical Spreadsheet workbook outputs, the reported value-based evaluator scores the complete SkillOpt

artifact at 10/12, whereas joint value-and-formula recalculation scores it at 8/12. Representation, cache materialization, or formula inspection can therefore change what the evaluator observes. This dual scoring is not identical in design to the Radar output-locked swap, but it reinforces the narrower point that score is a property of output plus measurement, not behavior alone.

Neither result proves semantic failure, exploitation, or benchmark-wide prevalence. To identify evolution-induced verifier dependence, one would additionally need matched no-skill, parent, and final outputs under both original and calibrated verifiers, and test whether the parent-to-final verifier gap increases.

Information-Integrity Boundary

This supporting check tests whether restricted answer or oracle information enters and controls the artifact. High-confidence direct-answer or oracle matches are absent from 86 SkillRevise and 85 SkillHone natural artifacts. SkillOpt conditions containing the feedback-derived answer-surface section follow incorrect metadata in 0/15 probes. In contrast, an unsafe hidden-oracle positive control changes probe success from 0/5 to 5/5 and leaves private-oracle or gold markers.

These checks show that the audit can detect direct exposure when it is deliberately introduced, while the natural artifacts do not support direct copying as a universal explanation. They do not prove the absence of implicit answer-following policies or all forms of leakage.

Observed score gains can arise at multiple lifecycle stages: feedback shapes candidate revisions, selection can amplify narrow improvements, adopted artifact components can mediate real behavior, and valid verifiers can disagree on fixed outputs. These mechanisms must be identified separately before a score gain is interpreted as skill learning.

From Reported Gains to Auditable Claims

Evidence Requirements and Claim Boundaries

The audit is not a scalar ladder from *safe* to *unsafe*. Its steps answer independent questions and impose distinct claim boundaries.

Audit step	Required test → maximum supported claim
Validity	Execution and verifier status plus assertion coverage → eligible observation
Retention	Frozen actual-parent/final fresh repeats → retained artifact effect
Behavioral scope	Entity, parameter, input, contract, or family variants → bounded controlled transfer
Evolution pathway	Feedback ablation, optimizer repeats, and gate provenance → pathway dependence or selection mediation
Measurement	Output-locked calibrated verifier comparison → measurement dependence
Information integrity	Provenance, wrong-metadata, and positive control → exploitation or confirmed leakage, when supported

Table 5: Evidence requirements and claim boundaries. The rows answer complementary questions rather than forming a scalar risk ladder.

A selected gain is weaker than a retained artifact effect; a retained effect on one instance is weaker than controlled transfer. Evolution-pathway and measurement claims remain meaningful even when retention is not established. Conversely, verifier-aware language or evaluator dependence must not be relabeled as leakage without restricted-information provenance and behavioral evidence.

Lifecycle Reporting Protocol

An auditable self-evolution report should preserve the following minimum evidence:

- **Before optimization:** parent artifact, task and split identity, failure pool, environment, and verifier version.
- **During revision:** optimizer-visible traces and feedback, candidate patches, optimizer samples, prompts, and usage.
- **At selection and adoption:** accepted and rejected artifacts, gate items, per-item decisions, and exact adopted artifact.
- **At deployment:** frozen artifact, actual-parent and no-skill conditions, fresh repeats, controlled variants, and output bytes.
- **At scoring:** verifier logs, assertion coverage, reference calibration, and any alternative verifier applied to the same output.
- **For cost:** separate optimizer, executor, verifier, and invalid-attempt ledgers.

The complete artifact appendix should additionally record hashes, commands, seeds, environment revisions, timeouts, and stopping decisions. This protocol turns a post-evolution score into an auditable lineage and limits the paper's language to the strongest matched contrast actually performed.

Related Work

Skill Evolution and Empirical Evaluation

Agent-skill research spans generation, revision, selection, retrieval, organization, and reinforcement learning. SkillsBench and SWE-Skills-Bench compare skill and no-skill utility across diverse or software-engineering tasks (Li et al. 2026; Han et al. 2026). SkillLearnBench evaluates continual skill generation (Zhong et al. 2026), while SkillRevise, SkillHone, SkillOpt, SkillRL, and CoEvoSkills propose different revision or learning loops (Liu et al. 2026b; Li and Hu 2026; Yang et al. 2026; Xia et al. 2026; Zhang et al. 2026b). Recent empirical work broadens the object of study: From Raw Experience to Skill Consumption separates experience generation, extraction, and consumption (Huang et al. 2026); How Well Do Agentic Skills Work in the Wild studies realistic skill retrieval (Liu et al. 2026a); SkillCoach evaluates process-level skill use (Zhu et al. 2026); and SkillJuror isolates the effect of skill organization (Chen et al. 2026).

These studies show that outcome-only evaluation is already recognized as incomplete. Our focus is narrower and complementary: when a persistent skill is revised and the

post-revision score rises, what evidence identifies a retained artifact effect rather than a selected or measurement-dependent endpoint? We therefore organize evidence by lifecycle contrast rather than by method score.

Evaluation and Attribution of Agent Improvement

Agent evaluation increasingly examines process quality, sequential adaptation, verifier reliability, and counterfactual behavior. Counterfactual Trace Auditing measures how a provided skill changes an agent trajectory (Zhou et al. 2026a). SEA-Eval evaluates self-evolving agents beyond isolated episodes (Jiang et al. 2026). Co-evolving verifier work and grader audits study interactions between the policy and its measurement signal (Zhang et al. 2026b,c). More broadly, reward hacking and test-set contamination research demonstrates that performance on a proxy or exposed test need not imply the intended capability (Amodei et al. 2016; Oren et al. 2023).

We combine these concerns within the lifecycle of persistent skill revision. The central distinction is between retention, behavioral scope, evolution-and-selection pathway, and output-level measurement dependence. Information-integrity checks play a supporting role: they bound direct answer or oracle exposure, but do not convert every verifier-aware adaptation into a leakage claim.

Limitations

Our evidence is mechanism-focused rather than a benchmark-wide prevalence study. The 24-task retention panel is purposive and contains only two fresh deployments per condition; it exposes instability but cannot estimate deployment rates. The controlled Invoice, Energy, SearchQA, Radar, Spreadsheet, and OfficeQA probes use small, heterogeneous task families rather than a fully crossed method-by-intervention design. SkillRevise, for example, lacks matched artifact surgery and output-locked measurement evidence of the same strength, while the strongest verifier swap is concentrated in the Radar family. We therefore do not claim that any observed mechanism is universal or that one method outperforms another.

Several conclusions are also bounded by observability. Output-locked comparisons identify measurement dependence but do not determine which verifier best captures intended behavior without an external specification. Artifact forensics and wrong-metadata controls can rule out simple direct exposure, but not implicit answer-following policies or every form of contamination. Finally, our scope is frozen-agent, feedback-driven systems with persistent external artifacts; parameter-updating self-evolution, multimodal skills, and online environments may exhibit different lifecycle dynamics. Confirmatory evidence will require a shared sampling plan that fully crosses actual parent and final artifacts, fresh repeats, controlled variants, and calibrated verifiers.

Conclusion

We study whether benchmark gains from self-evolving persistent skills correspond to retained behavioral improvement and, independently, which lifecycle mechanisms produce those gains. The variant comparison shows a substantial transfer advantage for SkillEvolver, including when all methods supply artifacts. Execution traces connect successful cases to reusable diagnostic steps and preservation of task constraints. Our audits show that revised skill artifacts can produce real behavioral changes, but selected gains can disappear under fresh deployment and the transfer established so far remains narrow. Feedback, optimizer variation, validation selection, artifact content, execution variability, and verifier choice can each shape the observed endpoint.

A higher benchmark score should therefore be reported only at the strongest level supported by matched evidence: a selected gain, a retained artifact effect, bounded controlled transfer, evolution-pathway dependence, or measurement dependence. We provide a lifecycle audit that links each claim to the required intervention and reporting evidence. Claims of transferable skill learning require frozen parent-to-final improvement that survives fresh deployment and predeclared task variation under valid measurement; weaker evidence should be reported more narrowly.

References

Amodei, D.; Olah, C.; Steinhardt, J.; Christiano, P.; Schulman, J.; and Mané, D. 2016. Concrete Problems in AI Safety. arXiv:1606.06565.

Chen, Z.; Guo, Z.; Huang, B.; Lu, B.; Lin, J.; Zhou, Y.; and Zhang, W. 2026. SkillJuror: Measuring How Agent Skill Organization Changes Runtime Behavior. arXiv:2606.11543.

Han, T.; Zhang, Y.; Song, W.; Fang, C.; Chen, Z.; Sun, Y.; and Hu, L. 2026. SWE-Skills-Bench: Do Agent Skills Actually Help in Real-World Software Engineering? arXiv:2603.15401.

Huang, Z.; Xu, J.; Yang, Y.; Gong, Z.; Yang, Q.; Tian, M.; Wang, X.; Lv, C.; Gao, X.; Dai, Q.; Liu, B.; Qiu, K.; Yang, X.; Chen, D.; Zheng, X.; and Luo, C. 2026. From Raw Experience to Skill Consumption: A Systematic Study of Model-Generated Agent Skills. arXiv:2605.23899.

Jiang, S.; Ma, L.; Hong, Z.; Wang, K.; Lu, Z.; Wang, T.; Chen, S.; Zhang, J.; Pan, T.; Li, W.; Liang, J.; and Xiao, Y. 2026. SEA-Eval: A Benchmark for Evaluating Self-Evolving Agents Beyond Episodic Assessment. arXiv:2604.08988.

Li, X.; Liu, Y.; Chen, W.; You, B.; Di, Z.; He, Y.; Zheng, S.; Choe, K. W.; Sun, J.; Wang, S.; Tao, C.; Li, B.; Zhao, X.; Geng, H.; Wu, X.; Zhou, J.; Chen, X.; Xing, H.; Li, Y.; Zeng, Q.; Wang, D.; Wang, Y.; Chaim, R. B.; Jiang, P.; Shen, H.; Kong, L.; Liu, X.; Wang, R.; Liu, X.; Li, J.; Lan, X.; Lin, Y.; Ye, W.; He, J.; Li, S.; Zhang, Y.; Gao, Y.; Li, Y.; Ma, Z.; Jing, L.; Wang, T.; Li, K.; Xue, Y.; Lyu, H.; He, Y.; Tian, Y.; Wu, S.; Wang, B.; Gao, Y.; Chen, B.; Liu, L.; Cheng, S.; Bao, J.; Tong, S.; Xu, S.; Zhuo, T. Y.; Ye, T.; Qi, Q.; Li, M.; Liao, L.; Tan, Z.; Shi, C.; Tang, X.; Tankasala, S.; Yuan, B.; Qian, Y.; Tu, J.; Wang, C.; Sun, Y.; Wang, W.; Taylor, A.; Yang, Z.; Guan, C.; Dong, Z.; Zhang, X.; Dillmann, S.; chung Lee, H.; and Song, D. 2026. SkillsBench: Benchmarking How Well Agent Skills Work Across Diverse Tasks. arXiv:2602.12670.

Li, Z.; and Hu, Y. 2026. SkillHone: A Harness for Continual Agent Skill Evolution Through Persistent Decision History. arXiv:2606.08671.

Liu, Y.; Ji, J.; An, L.; Jaakkola, T.; Zhang, Y.; and Chang, S. 2026a. How Well Do Agentic Skills Work in the Wild: Benchmarking LLM Skill Usage in Realistic Settings. arXiv:2604.04323.

Liu, Y.; Su, Z.; Xie, L.; Zhang, Y.; Zong, Q.; Guo, J.; Xie, Z.; Ji, Y.; Yim, Y.; Luo, H.; Ren, X.; Ruan, C.; Li, H.; and Song, Y. 2026b. SkillRevise: Improving LLM-Authored Agent Skills via Trace-Conditioned Skill Revision. arXiv:2606.01139.

Oren, Y.; Meister, N.; Chatterji, N.; Ladhak, F.; and Hashimoto, T. B. 2023. Proving Test Set Contamination in Black Box Language Models. arXiv:2310.17623.

Xia, P.; Chen, J.; Wang, H.; Liu, J.; Zeng, K.; Wang, Y.; Han, S.; Zhou, Y.; Zhao, X.; Chen, H.; Zheng, Z.; Xie, C.; and Yao, H. 2026. SkillRL: Evolving Agents via Recursive Skill-Augmented Reinforcement Learning. arXiv:2602.08234.

Yang, Y.; Gong, Z.; Huang, W.; Yang, Q.; Zhou, Z.; Huang, Z.; Li, Y.; Gao, X.; Dai, Q.; Liu, B.; Qiu, K.; Yang, Y.; Chen, D.; Yang, X.; and Luo, C. 2026. SkillOpt: Executive Strategy for Self-Evolving Agent Skills. arXiv:2605.23904.

Zhang, G.; Zhu, E.; Zhou, J.; Jia, C.; and Wang, H. 2026a. SkillEvolver: Skill Learning as a Meta-Skill. arXiv:2605.10500.

Zhang, H.; Fan, S.; Zou, H. P.; Chen, Y.; Wang, Z.; Zhou, J.; Li, C.; Huang, W.-C.; Yao, Y.; Zheng, K.; Liu, X.; Li, X.; and Yu, P. S. 2026b. CoEvoSkills: Self-Evolving Agent Skills via Co-Evolutionary Verification. arXiv:2604.01687.

Zhang, X.; Wang, G.; Cui, Y.; Li, Z.; Qiu, W.; Zhu, B.; and He, P. 2026c. Who Grades the Grader? Co-Evolving Evaluation Metrics and Skills for Self-Improving LLM Agents. arXiv:2607.12790.

Zhong, S.; Lu, Y.; Ning, J.; Wan, Y.; Feng, L.; Ao, Y.; Ribeiro, L. F. R.; Dreyer, M.; Ammirati, S.; and Xiong, C. 2026. SkillLearnBench: Benchmarking Continual Learning Methods for Agent Skill Generation on Real-World Tasks. arXiv:2604.20087.

Zhou, X.; Liu, J.; Li, L.; Rossi, R. A.; and Hu, X. 2026a. Counterfactual Trace Auditing of LLM Agent Skills. arXiv:2605.11946.

Zhou, Y.; Dong, Z.; Wang, Z.; Jin, C.; Zhao, S.; Guo, B.; Gu, D.; Zhang, L.; Zhou, M.; and Metaxas, D. N. 2026b. Evidence Over Plans: Online Trajectory Verification for Skill Distillation. arXiv:2605.09192.

Zhu, J.; Mao, K.; Guo, Y.; He, D.; Xu, S.; Gu, S.; and Yue, Y. 2026. SkillCoach: Self-Evolving Rubrics for Evaluating and Enhancing Agentic Skill-Use. arXiv:2607.01874.